MoodTune AI: Song Emotion Analysis and Emotion-Based Music Recommendation Using DistilBERT

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Abstract

Music is an important medium for expressing emotions, and users often prefer songs that match their emotional state. Most existing music recommendation systems primarily rely on listening history, song genre, artist preference, or popularity, while giving limited importance to the emotional meaning of song lyrics. This project proposes an AI-based Song Emotion Analysis and Emotion-Based Music Recommendation System that utilizes Natural Language Processing (NLP) and a pre-trained DistilBERT model to analyze the emotional content of song lyrics. The system accepts a song name as input, retrieves the corresponding lyrics from a lyrics dataset, and processes the text using DistilBERT to classify the dominant emotion into categories such as Happy, Sad, Love, Anger, Fear, and Motivational. Based on the predicted emotion, the system recommends songs with similar emotional characteristics, providing a more meaningful and personalized music recommendation experience. The proposed approach leverages the contextual understanding capability of DistilBERT to improve emotion classification compared to traditional keyword-based methods. The system is designed using open-source datasets and modern NLP techniques, making it scalable and suitable for real-world applications. Future enhancements include multilingual lyric analysis, integration of audio features, and compatibility with online music streaming platforms.

Keywords: Natural Language Processing (NLP); DistilBERT; Song Emotion Analysis; Music Recommendation; Emotion Classification; Artificial Intelligence.

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